

Zihong Lu

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Education

Harbin Institute of Technology, Shenzhen	2024/09 – Present
M.Eng. in Control Science and Engineering, School of Intelligent Science and Engineering	Shenzhen, China
Harbin Institute of Technology, Shenzhen	2020/09 – 2024/06
B.Eng. in Automation, School of Intelligent Science and Engineering	Shenzhen, China

Experience

STAR Robotics Lab, Southern University of Science and Technology	2024/09 – Present
Visiting Researcher	
• Conducted research on UAV autonomy, focusing on autonomous exploration, motion planning, and reinforcement learning control for agile aerial robots.	
Multi-Agent System Lab, Harbin Institute of Technology, Shenzhen	2024/09 – Present
Graduate Student Researcher	
• Participated in ongoing research on UAV perception, planning, and swarm situational awareness, including dynamic obstacle perception and trajectory planner integration. Collaborated with fellow lab members in multiple UAV-related competitions.	
Complex Terrain Robotics Group, Tsinghua Shenzhen International Graduate School	2021/10 – 2022/03
Undergraduate Student Researcher	
• Developed uneven-terrain planning methods and assisted code optimization for dynamic obstacle avoidance with model predictive control and control barrier functions for safe mobile robot navigation.	
Electronic Innovation and Design Community, Harbin Institute of Technology, Shenzhen	2021/12 – 2025/04
Assistant Manager & Teaching Assistant	
• Managed daily laboratory operations, maintained equipment, and supported the organization of student competitions.	
• Provided technical support for laser cutting, 3D printing, and embedded systems development.	

Research Interests

My research interests focus on autonomous robotics, especially UAV autonomous exploration, motion planning, and reinforcement learning control, with broader interests in humanoid robotics and vision-language navigation.

Publications

PerchRL: Vision-Based Agile Perching on Inclined Platforms under Rapid and Irregular Motion
Zihong Lu [*] , Zongzhuo Liu [*] , Huaxu Li, Jinqiang Cui, Jie Mei, Youmin Gong, U Kei Cheang, Boyu Zhou [†]
Under Review, CoRL 2026. [Paper] [Video]
Perception-aware Planning for Quadrotor Flight in Unknown and Feature-limited Environments
Chenxin Yu [*] , Zihong Lu [*] , Jie Mei, Boyu Zhou [†]
2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025). [Paper] [Video] [Code] - Oral
Dynamic Control Barrier Function-based Model Predictive Control to Safety-Critical Obstacle-Avoidance of Mobile Robot
Zhuozhu Jian [*] , Zihong Yan [*] , Xuanang Lei, Zihong Lu, Bin Lan [†] , Xueqian Wang [†] , Bin Liang
2023 IEEE International Conference on Robotics and Automation (ICRA 2023). [Paper] [Video] [Code]
PUTN: A Plane-fitting based Uneven Terrain Navigation Framework
Zhuozhu Jian [*] , Zihong Lu [*] , Xiao Zhou, Bin Lan, Anxing Xiao, Xueqian Wang [†] , Bin Liang
2022 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2022). [Paper] [Video] [Code]

Technical Skills

- Programming Languages:** C/C++, Python, MATLAB
- Robotics Tools:** Ubuntu, ROS1/ROS2, Gazebo, AirSim, Isaac Sim, Unity, Qt, AutoCAD, SolidWorks, PX4
- Theoretical Knowledge:** Motion Planning, Reinforcement Learning, Autonomous Navigation, Autonomous Exploration, Model Predictive Control (MPC)
- Libraries and Frameworks:** Eigen, Ceres, OpenCV, PyTorch, TensorRT

Selected Honors and Awards

Third Place - DJI RoboMaster 2023-2024 University AI Challenge	Apr. 2024
National Silver Award - National College Students Engineering Practice and Innovation Ability Competition	Nov. 2021
National First Prize - National Undergraduate Electronics Design Contest	Dec. 2021
National Second Prize - Lanqiao Cup Electronic Design Contest, MCU Design and Development	Jun. 2022

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